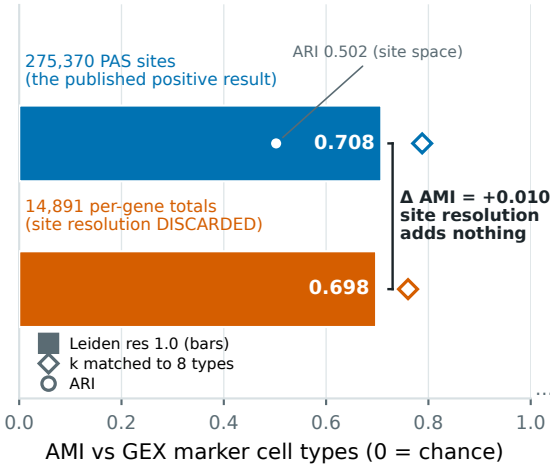


Fig S10 | PAS-profile clustering: the dropped claim, stated honestly

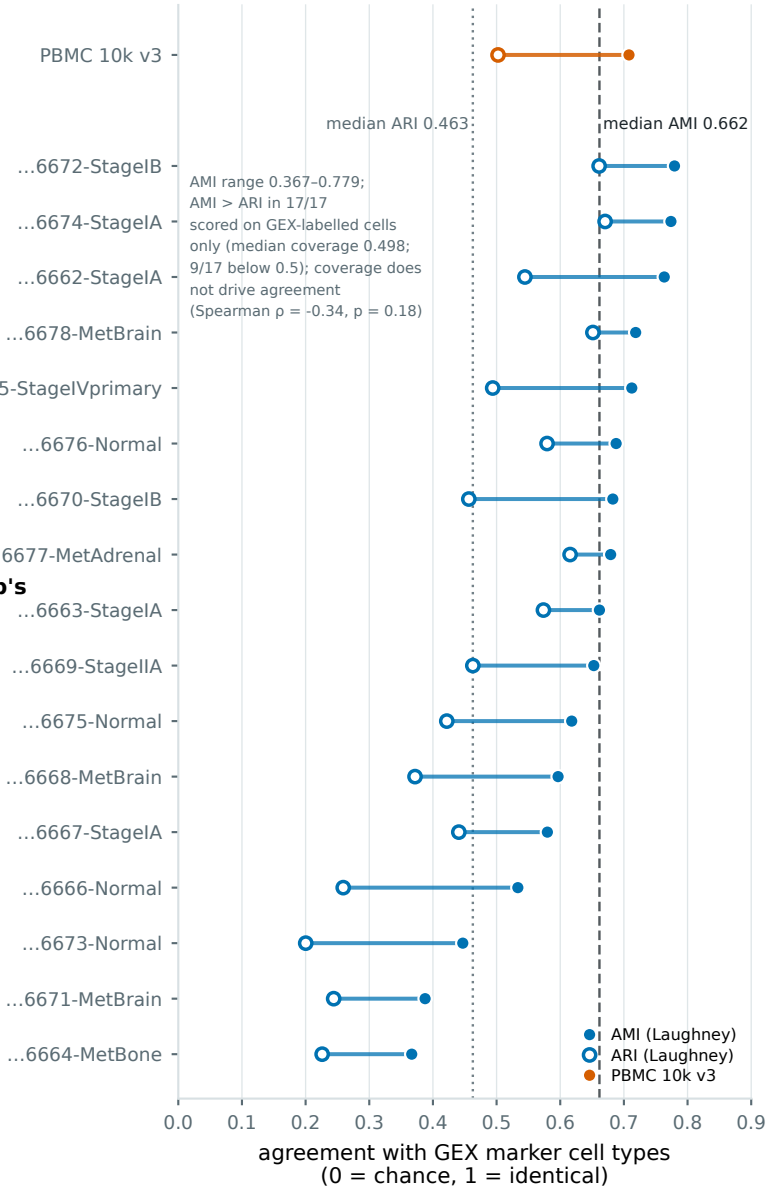
Cells cluster on poly(A)-site count profiles (TF-IDF + LSI + Leiden) and recover GEX cell types — but collapsing the sites to per-gene totals recovers them just as well: the signal is 3'-end gene expression re-encoded, not isoform choice. Offered only as an optional utility for label-free exploration when no GEX labels exist (01 §54).

Reference partition = marker-signature labels assigned per GEX Leiden cluster — an automated partition, not curated ground truth.
Features are per-site counts (TF-IDF + LSI), which carry gene abundance.

a The ablation that dropped the claim (PBMC)



b The (dropped) recovery replicates in 17/17 Laughney samples — but gene totals suffice (a)



c One knob governs granularity (retired sweep's sole durable finding)

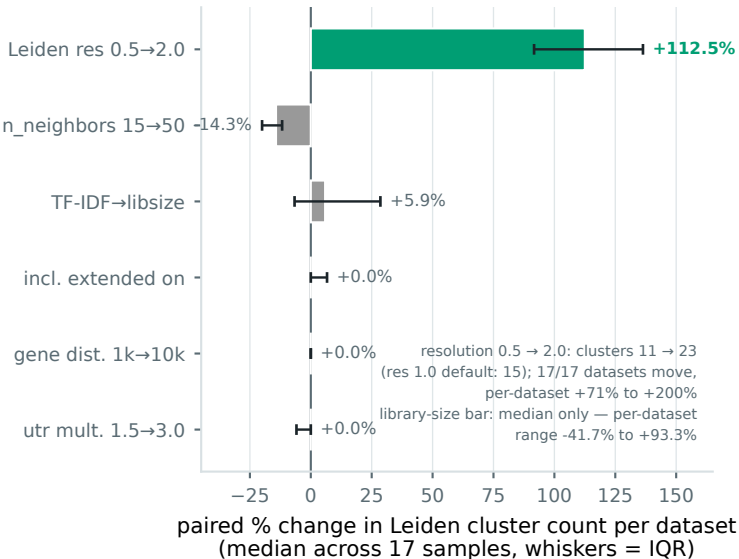


Fig S10 | The dropped clustering claim, stated honestly (optional utility only). (a) PBMC 10k v3: clustering on all 275,370 PAS-site counts recovers GEX marker cell types (AMI 0.708, ARI 0.502) — but summing the same counts to 14,891 per-gene totals, discarding all site-level resolution, recovers them just as well (AMI 0.6978; $\Delta +0.0102$; k-matched-to-8-types clusterings 0.7878 vs 0.7602). The site resolution adds nothing, so the recovery is 3'-end expression signal re-encoded, not isoform choice (01 §54). The isoform-only USAGE-fraction spaces of the same experiment are NOT shown: they were computed on mis-keyed matrices and their re-run is pending; the ablation verdict does not depend on them (01 §54). (b) The (dropped) positive result replicates: 17/17 Laughney samples, median AMI 0.662 (range 0.367-0.779), median ARI 0.463 (range 0.200-0.670), AMI > ARI in 17/17; the reference partition is marker-derived, not curated truth, and concordance is scored on GEX-labelled cells only (median coverage 0.498, 9/17 below 0.5; coverage does not drive agreement, Spearman $\rho = -0.34$, $p = 0.18$). (c) The retired parameter sweep's one durable finding: Leiden resolution dominates cluster count (+112.5% median, 0.5 to 2.0, 17/17 datasets; clusters 11 to 23) while every annotation-trim knob is at least 8x smaller. BINDING LABEL (results/paramsweep/VERDICTS.md 1): --min-pas-prominence is PROVEN byte-identically inert on the lambda strategies, so no prominence arm of the retired sweep could ever have been informative; the sweep ran on the pre-clip-seeded lambda_gradient cohort run and only this clustering-knob finding survives. Sources: pbmc_novelty.tsv (verified ablation rows), B2_gex_celltyping/concordance.csv (verified 05 R2), parameter_sweep.tsv (verified 05 R3).